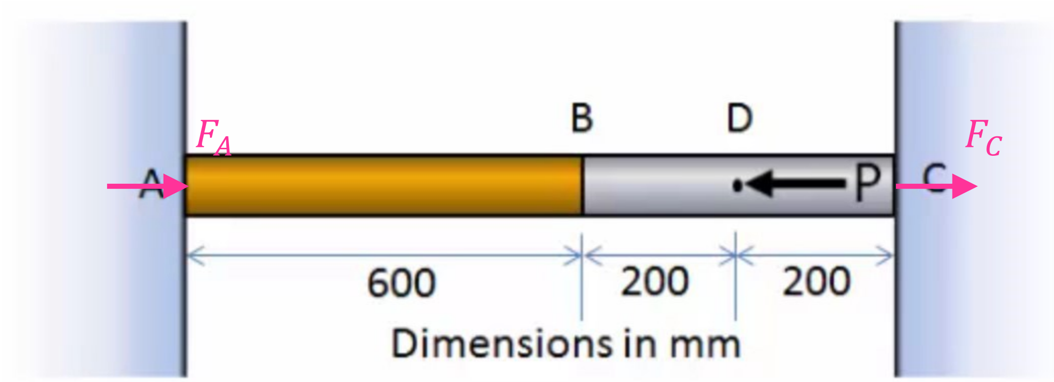
Big Extra Exercise for Solid Mechanics (EMCH260, Fall 2021)

1. The bar has constant diameter of 100 mm and is composed of two different materials. The two ends *A* and *C* are fixed to the wall. An external load ***P*** of 380 kN is applied at point *D*. The modulus of elasticity of part *AB* and *BC* is 80 GPa and 200 GPa, respectively. Determine the reactions at ends *A* and *C*. (**10'** in total)



Solutions:

Step 1: Equilibrium

Assuming the resultant reactions at the two ends *A* and *C* as shown in the figure, for the entire bar, the free-body diagram (force equation of equilibrium) gives:

$\sum F_x = 0 \rightarrow$

$F_A + F_C = P = 380 \text{ kN}$ **(1)**

Step 2: Compatibility Condition

The total strain or displacement of the entire bar (from *A* to *C*) is zero, i.e.

$\delta_{A/C} = 0 \rightarrow$

$\delta_{AB} + \delta_{BD} + \delta_{DC} = 0$ **(2)**

Step 3: Load-Displacement Relations

The elongation / compression of the bar can be obtained by the force (related to stress) and modulus of elasticity:

$$\delta_{AB} = -\frac{F_A L_{AB}}{E_{AB} A_{AB}}$$

$$\delta_{BD} = -\frac{F_A L_{BD}}{E_{BD} A_{BD}}$$

$$\delta_{DC} = +\frac{F_C L_{DC}}{E_{DC} A_{DC}}$$

Substituting them into Eq. (2), we get

$$-\frac{F_A L_{AB}}{E_{AB} A_{AB}} - \frac{F_A L_{BD}}{E_{BD} A_{BD}} + \frac{F_C L_{DC}}{E_{DC} A_{DC}} = 0$$

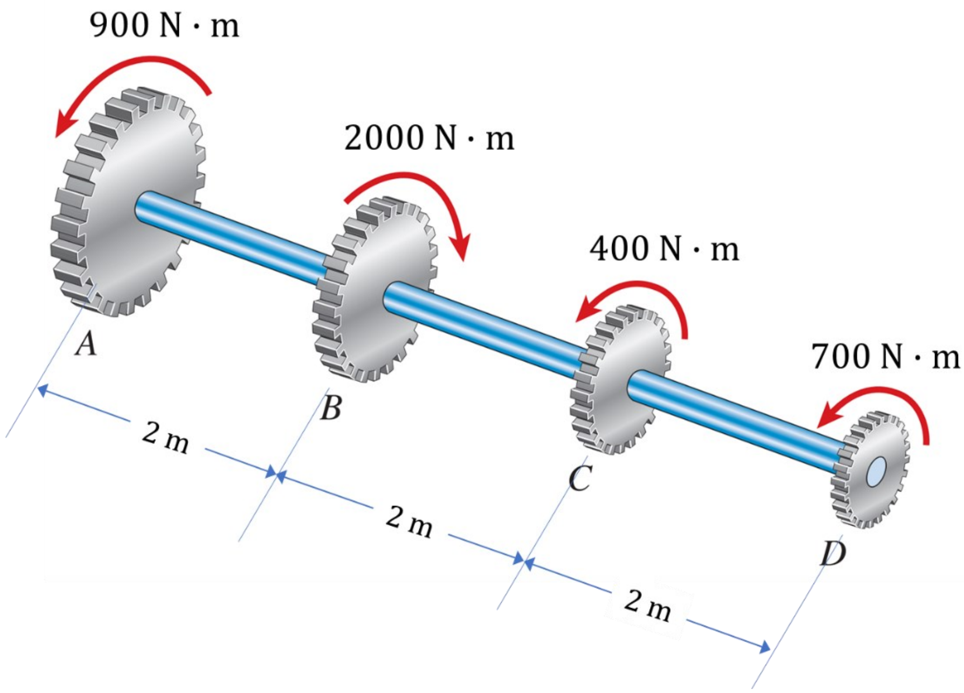
→ $\frac{F_C}{200} = \frac{3F_A}{80} + \frac{F_A}{200}$

→ $F_C = 8.5F_A$ **(3)**

Combining Eqs. (1) and (3), we obtain:

$$F_A = \boxed{\mathbf{40\ kN}}$$
$$F_C = \boxed{\mathbf{340\ kN}}$$

2. The allowable shear stress in the shaft is 70 MPa and the allowable angle of twist of end *A* relative to end *D* is 0.01 rad. What is the required minimum diameter d of the shaft if it has a solid cross section? The shear modulus of elasticity of the shaft is 80 GPa. (**15'** in total)



Solutions:

Section the shaft between *AB*, *BC*, and *CD* and apply the Free-Body-Diagram to the left or right segment to obtain the internal resultant torque (note: use the right-hand rule to determine the sign of internal torques):

$T_{AB} = -900\ \mathrm{N \cdot m}$

$T_{BC} = 2000 - 900 = +1100\ \mathrm{N \cdot m}$

$T_{CD} = +700\ \mathrm{N \cdot m}$

According to the torsion formula, the **maximum shear stress** will occur in *BC* region. Thus,

$$(\tau_{\max})_{BC} = \frac{T_{BC}c}{J} = \frac{1100 \times d/2}{\frac{\pi}{2} \times (d/2)^4} \leq \tau_{allow} = 70\ \mathrm{MPa}$$

$$d \geq \left(\frac{16 \times 1100}{\pi \times 70 \times 10^6}\right)^{1/3}$$

$d \geq 0.043\ \mathrm{m}$ **(1)**

The angle of twist of end *A* relative to end *D* is

$$\phi_{AD} = \phi_{AB} + \phi_{BC} + \phi_{CD} = \frac{T_{AB}L_{AB}}{JG} + \frac{T_{BC}L_{BC}}{JG} + \frac{T_{CD}L_{CD}}{JG}$$

$$= -\frac{900 \times 2}{\frac{\pi}{2} \times \left(\frac{d}{2}\right)^4 \times 80 \times 10^9} + \frac{1100 \times 2}{\frac{\pi}{2} \times \left(\frac{d}{2}\right)^4 \times 80 \times 10^9} + \frac{700 \times 2}{\frac{\pi}{2} \times \left(\frac{d}{2}\right)^4 \times 80 \times 10^9}$$

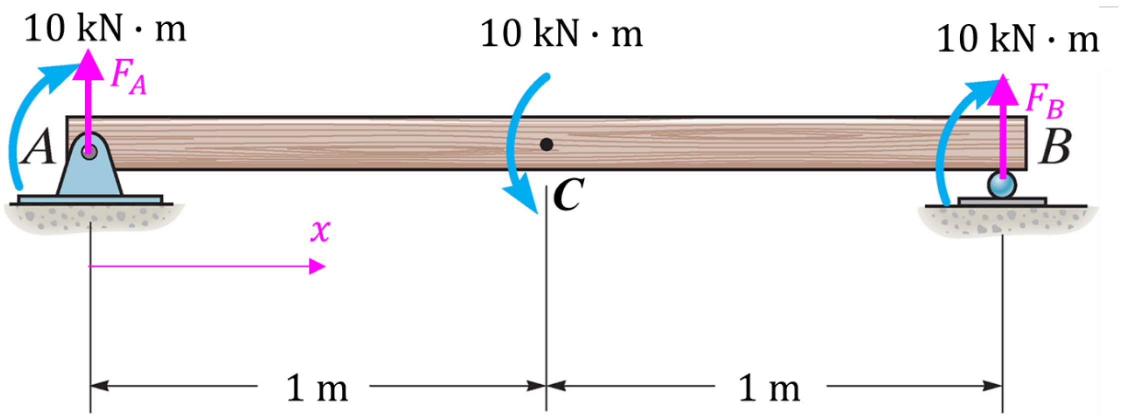
$$= \frac{1800}{\frac{\pi}{2} \times \left(\frac{d}{2}\right)^4 \times 80 \times 10^9}$$

To meet the requirement of $\phi_{AD} \leq 0.01$, we get

$d \geq 0.069$ m **(2)**

Combining Eqs. (1) and (2), we know that the required minimum diameter of the shaft is $\boxed{\mathbf{0.069\ m}.}$

3. Determine the absolute maximum normal stress in the beam with external loading shown below. The beam has a uniform square cross-section with lateral size $a = 0.1$ m. Note: the shear and moment diagrams can be calculated either by section method or graphical method. (**15'** in total)



Solutions:

Use graphical method to calculate the shear and moment diagrams

Step 1: Support Reactions

Applying free body diagram to the entire beam, the force equation of equilibrium gives:

$\sum F_y = 0 \quad \rightarrow \quad F_A + F_B = 0$

$\sum M_A = 0 \quad \rightarrow \quad F_B \times 2 + 10000 = 10000 + 10000 = 20000$

Solving the above equations yields:

$F_A = -5000$ N

$F_B = 5000$ N

Step 2: Shear Diagram

a) Determine the shear force at the left end:

$x = 0,\ \ V = F_A = -5000$ N

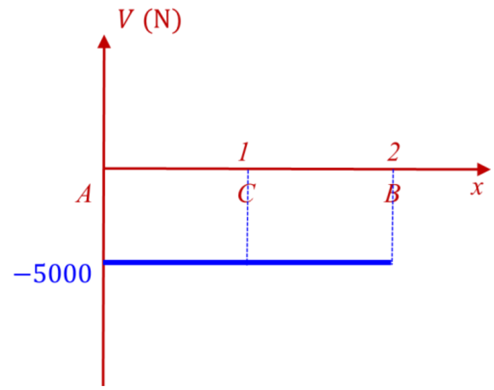
b) Determine **V** diagram according to $\frac{dV}{dx} = w$ for $\underline{AB}$ region (the entire beam) *since there is no external concentrated or distributed forces*:

$$\frac{dV}{dx} = w = 0$$

$\rightarrow\ V = C_1$

The constant C_1 can be determined by applying the "boundary" condition ($x = 0,\ \ V = -5000$ N). Thus, $C_1 = -5000$ N and $V = -5000\ \ (0 \le x \le 2\ \text{m})$.

Step 3: Moment Diagram



a) Determine **M** at left end first (the origin A of x-axis):

$x = 0, \ \ M = 10000 \text{ N} \cdot \text{m}$

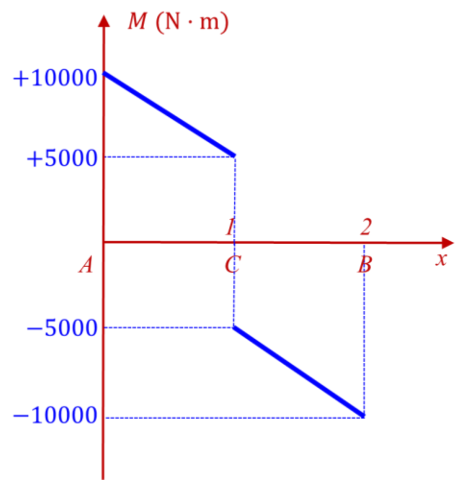
b) Determine **M** diagram according to $\frac{dM}{dx} = V$ for the $\underline{AC}$ region:

$$\frac{dM}{dx} = V = -5000$$

$\rightarrow \ M = -5000x + C_2$

The constant C_2 can be determined by applying the "boundary" condition ($x = 0, \ \ M = 10000 \text{ N} \cdot \text{m}$). Thus, $C_2 = 10000 \text{ N} \cdot \text{m}$ and $M = -5000x + 10000 \ \ (0 \le x \le 1 \text{ m})$.

c) At point C, the external bending moment (counterclockwise) will induce a decrease of internal moment by $10000 \text{ N} \cdot \text{m}$. Therefore, the internal moment will decrease from $+5000 \text{ N} \cdot \text{m}$ to $-5000 \text{ N} \cdot \text{m}$. See moment diagram for details.



d) Determine **M** diagram according to $\frac{dM}{dx} = V$ for $\underline{CB}$ region:

$$\frac{dM}{dx} = V = -5000$$

$\rightarrow \ M = -5000x + C_3$

The constant C_3 can be determined by applying the "**continuity**" condition at last point C after considering jump ($x = 1, \ \ M = -5000 \text{ N} \cdot \text{m}$). Thus, $C_3 = 0$ and $M = -5000x \ (1 \text{ m} \le x \le 2 \text{ m})$.

Therefore, the maximum moment throughout the entire beam is $M_{\max} = 10 \text{ kN} \cdot \text{m}$ at $x = 0$ and 2 m.

Step 4: Section Property

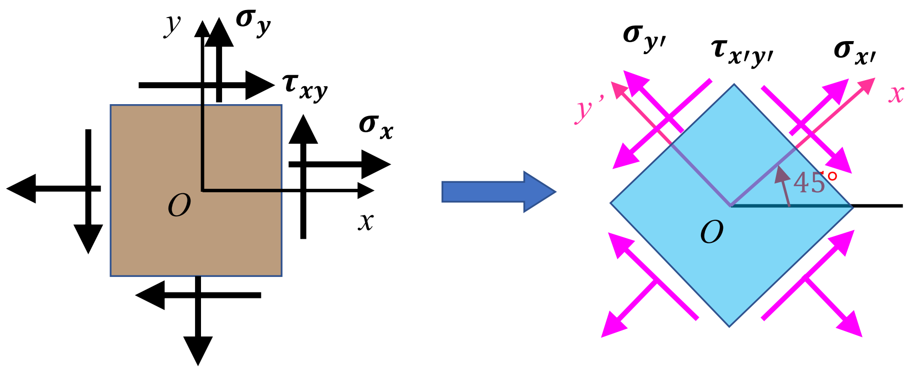
The moment of inertia I of sectional area about the neutral axis is (refer to the inside cover of the textbook for rectangular cross-section):

$$I = \frac{1}{12}bh^3 = \frac{1}{12} \times 0.1 \times 0.1^3 = 8.3333 \times 10^{-6} \text{ m}^4$$

Step 4: Maximum Normal Stress

$$\sigma_{\max} = \frac{M_{\max}c}{I} = \frac{10000 \times 0.05}{8.3333 \times 10^{-6}} = \boxed{\textbf{60 MPa}}$$

4. For a plane stress state, in the original *Oxy* system we know the component $\sigma_x = 120$ MPa and $\tau_{xy} = 60$ MPa. After the element (see the left figure) is rotated counterclockwise by 45°, we know the stress component $\tau_{x'y'} = -60$ MPa in the new *Ox'y'* system (see right figure). Determine the two principal stresses and the absolute maximum shear stress ($\tau_{\max}^{\text{abs}}$). (**10'** in total)



Solutions:

The reference point on the Mohr's circle is $A(120, 60)$. After the element is rotated counterclockwise by 45°, accordingly, the point A is rotated to point $B(\sigma_{x'}, \tau_{x'y'})$. We label point A and B on the circle.

According to the geometry and property of the Mohr's circle, the angle $\angle ACB = 90°$ and

$\tau_{xy} = 60$ MPa , $\tau_{x'y'} = -\tau_{xy} = -60$ MPa. Thus, $AD = DB = 60$ MPa.

In ΔADC, since $CD = DA = 60$ MPa and $\angle ADC = 90°$, we have:

$R = \sqrt{2} \times 60 = 84.85$ MPa

The center of the Mohr's circle can be determined as

$\sigma_C = 120 - 60 = 60$ MPa

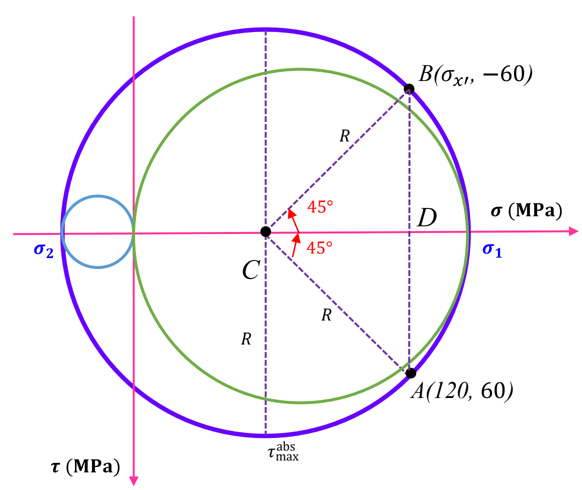
The two principal stresses are then determined:

$\sigma_1 = \sigma_C + R = 60 + 84.85 = \boxed{\mathbf{144.85\ MPa}}$

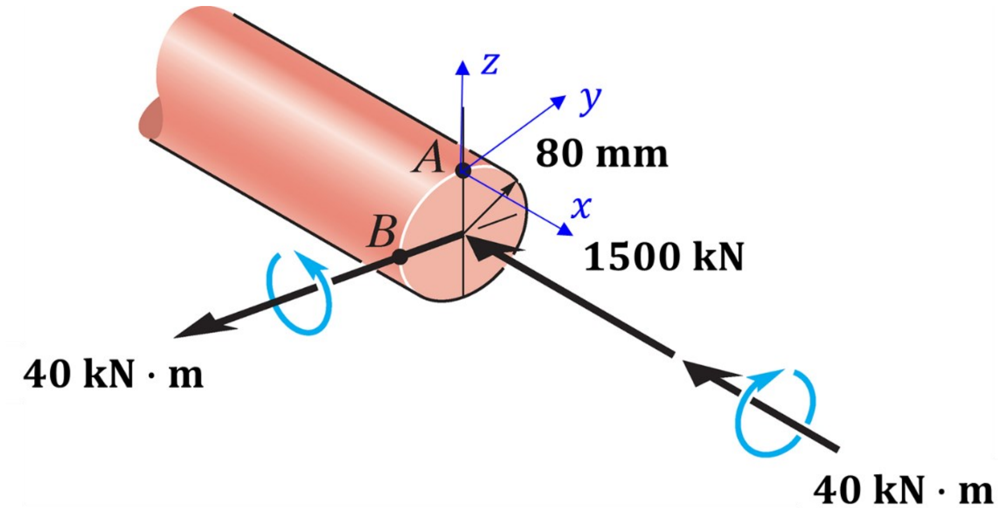
$\sigma_2 = \sigma_C - R = 60 - 84.85 = \boxed{\mathbf{-24.85\ MPa}}$

The absolute maximum shear stress is:

$\tau_{\max}^{\text{abs}} = R = \boxed{\mathbf{84.85\ MPa}}$



5. The internal loadings at a cross section through the shaft consist of an axial force of 1500 kN, a bending moment of 40 kN · m, and a torsional moment of 40 kN · m. The radius of the shaft is 80 mm. Determine the principal stresses at the surface point *A* and the <u>absolute maximum</u> shear stress at this point. (**15'** in total)



<u>Solutions:</u>

<u>Step 1: Determine Stress State</u>

At point *A*, the normal stress is induced by both axial load and bending moment, both of which are compressive.

$$\sigma_x = -\frac{1500 \times 1000}{\pi \times 0.08^2} - \frac{40000 \times 0.08}{\frac{\pi}{4} \times 0.08^4} = -174.1 \text{ MPa}$$

The shear stress is induced by the torsional moment.

$$\tau_{xy} = \frac{Tc}{J} = \frac{40000 \times 0.08}{\frac{\pi}{2} \times 0.08^4} = 49.7 \text{ MPa}$$

$$\sigma_y = 0$$

<u>Step 2: Construction of the Mohr's Circle</u>

Given $\sigma_x = -174.1$ MPa, $\sigma_y = 0$, $\tau_{xy} = 49.7$ MPa, the center of the circle *C* is located on the σ-axis, at the point $\sigma_{\text{avg}} = \frac{\sigma_x + \sigma_y}{2} = -87.05$ MPa. The point $C(-87.05, 0)$ and the reference point $A(-174.1, 49.7)$ are labeled. Then, the radius of the circle is $R = \sqrt{(-87.05 - (-174.1))^2 + 49.7^2} = 100.24$, and the Mohr's circle is constructed.

<u>Step 3: Principal Stresses</u>

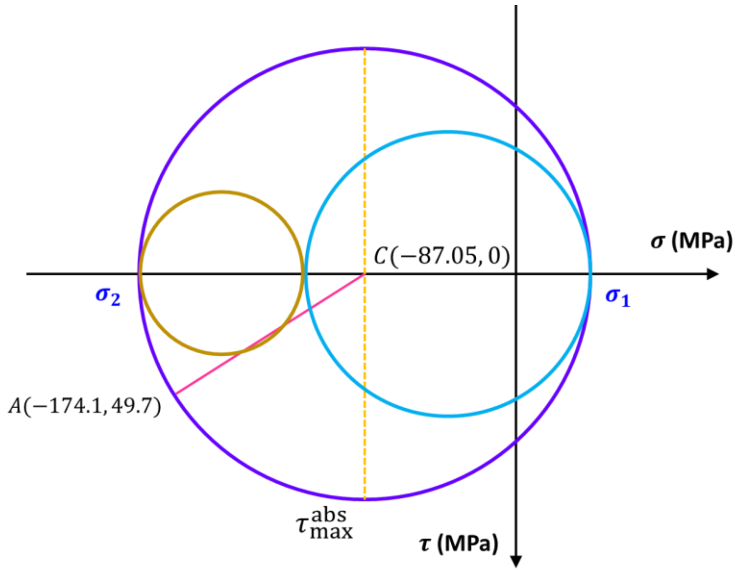
$$\sigma_1 = -87.05 + 100.24 = \boxed{\mathbf{13.19\ MPa}}$$

$$\sigma_2 = -87.05 - 100.24 = \boxed{\mathbf{-187.29\ MPa}}$$

Step 4: Absolute Maximum Shear Stress

Since σ_1 and σ_2 have opposite signs, applying Eq. (9-14) on textbook page 482 we have

$$\tau_{\max}^{\text{abs}} = \frac{\sigma_1 - \sigma_2}{2} = R = \boxed{\mathbf{100.24\ MPa}}$$

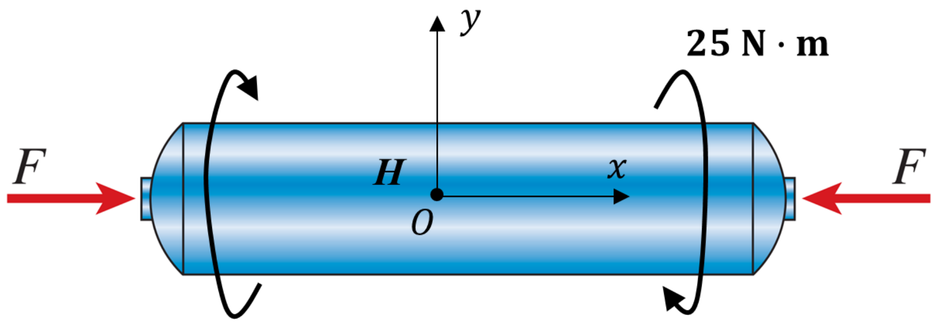


6. A thin-walled cylindrical pressure vessel of inner radius $r = 0.6$ cm and a thickness of $t = 0.06$ cm is subjected simultaneously to internal gas pressure $p = 3.5$ MPa, a compressive force $F = 800$ N acting at the ends, and a torsional load of 25 N · m (see figure). Determine the following quantities (**15'** in total):

(a) the principal stresses at a point *H* on the surface of the vessel and the absolute maximum shear stress; (**10'**)

(b) the normal strain along x-axis (ε_x) and the shear strain (γ_{xy}) at the same point *H*. See the established *Oxy* coordinate system. The Young's modulus of the material is 200 GPa and the Poisson's ratio is 0.3. (**5'**)

Hint: the cross-section of the vessel can be regarded as a hollow tube and the cross-section area can be calculated as $\pi(0.0066^2 - 0.006^2)$. You can do the same treatment for the polar moment inertia.

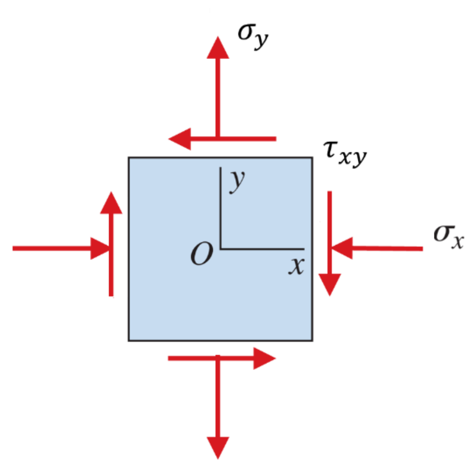


Solutions:

(a) In this gas filled vessel, in addition to the two non-equal normal stresses (σ_n and σ_a) in the two perpendicular directions (σ_n in the longitudinal (x-) direction and σ_a in the hoop (y-) direction), there is shear stress (τ_{xy}) induced by the external torque and compressive normal stress (denoted by σ_F) induced by the external compressive force.

Cross-sectional area:

$$A = \pi(0.0066^2 - 0.006^2) = 2.375 \times 10^{-5}\ \text{m}^2$$



Polar moment of inertia:

$$J = \frac{\pi}{2}(0.0066^4 - 0.006^4) = 9.448 \times 10^{-10}\ \text{m}^4$$

$$\sigma_n = \frac{pr}{2t} = \frac{3.5 \times 10^6 \times 0.6}{2 \times 0.06} = 17.5\ \text{MPa}$$

$$\sigma_a = \frac{pr}{t} = \frac{3.5 \times 10^6 \times 0.6}{0.06} = 35\ \text{MPa}$$

$$\tau_{xy} = -\frac{Tc}{J} = -\frac{25 \times 0.0066}{9.448 \times 10^{-10}} = -174.6\ \text{MPa}$$

$\sigma_F = \frac{F}{A} = \frac{800}{2.375 \times 10^{-5}} = 33.68 \text{ MPa}$

Then, in the original coordination system Oxy the overall stress state for point *H* can be expressed as

$\sigma_x = \sigma_n - \sigma_F = 17.5 - 33.68 = -16.18 \text{ MPa}$ (negative sign means compressive stress)

$\sigma_y = \sigma_a = 35 \text{ MPa}$

$\tau_{xy} = -174.6 \text{ MPa}$

Given these values, the center of the Mohr's circle *C* is located at the point *C*(9.41, 0) and the reference point *A*(-16.18, -174.6) are labeled. Then, the radius of the circle is $R = \sqrt{(9.41 + 16.18)^2 + 174.6^2} = 176.47$, and the circle is constructed (see figure).

The principal stresses are:

$\sigma_1 = 9.41 + 176.47 = \boxed{\mathbf{185.88 \text{ MPa}}}$

$\sigma_2 = 9.41 - 176.47 = \boxed{\mathbf{-167.06 \text{ MPa}}}$

The absolute maximum shear stress is:

$\tau_{\max}^{\text{abs}} = R = \boxed{\mathbf{176.47 \text{ MPa}}}$

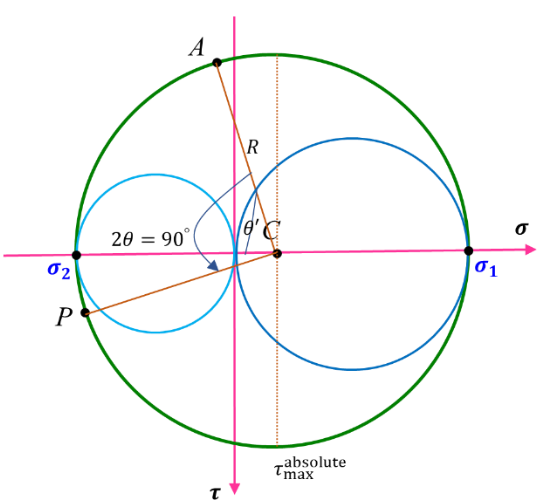
(b) In the *Oxy* coordination system, using the generalized Hooke's law, we obtain

$\varepsilon_x = \frac{\sigma_x - \nu(\sigma_y + \sigma_z)}{E} = \frac{-16.18 - 0.3 \times (35 + 0)}{200 \times 10^9} \times 10^6$

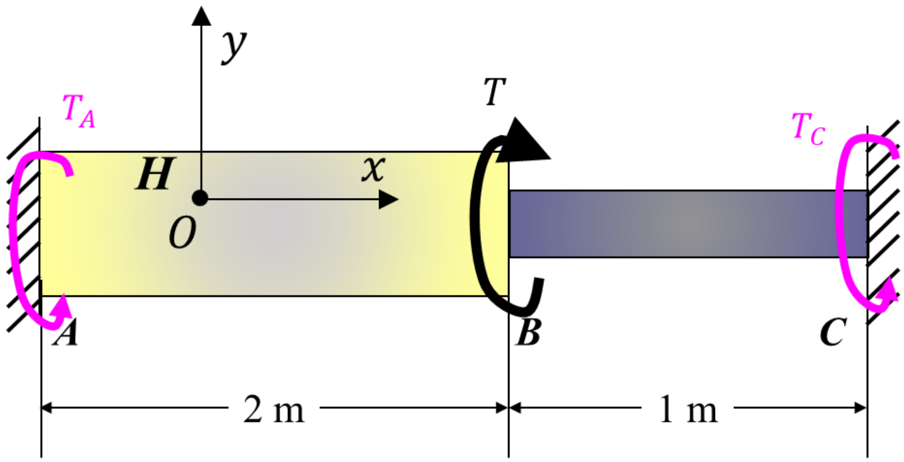
$= \boxed{\mathbf{-1.334 \times 10^{-4}}}$

Using relationship between material properties, the shear modulus $G = \frac{E}{2(1+\nu)} = 76.92 \text{ GPa}$ and

$\gamma_{xy} = \frac{\tau_{xy}}{G} = \frac{-174.6 \times 10^6}{76.92 \times 10^9} = \boxed{\mathbf{-2.270 \times 10^{-3} \text{ rad}}}$



7. The compound shaft is composed of bronze cylindrical segment AB and steel cylindrical segment BC. The two ends of the compound shaft are fixed to rigid supports. The external torque (T) is applied at B. The shear modulus of segment AB and BC is 40 GPa and 80 GPa, respectively. The length of the segment AB and BC is 2 m and 1 m, respectively. The diameter of the segment AB and BC is 0.05 m and 0.025 m, respectively. Determine the external torque T applied to point B so that one of the principal stresses for a point H on the surface of the segment AB is -100 MPa. (**20'** in total)



Solutions:

Assuming the reactions at the fixed ends A and C as T_A and T_C, respectively, the free-body diagram gives:

$T_A + T_C = T$ (1)

This is a torque related indeterminate problem. The compatibility condition is the total relative twist angle between A and C end is zero, i.e.

$\phi_{A/C} = 0$ →

$\phi_{AB} + \phi_{BC} = 0$ (2)

According to the twist angle formula (denote the diameter of segment AB and BC as D and d, respectively):

$\phi_{AB} = -\frac{T_A L_{AB}}{J_{AB} G_{AB}} = -\frac{T_A L_{AB}}{\frac{\pi}{2}(D/2)^4 G_{AB}}$ (3a)

$\phi_{BC} = \frac{T_C L_{BC}}{J_{BC} G_{BC}} = \frac{T_C L_{BC}}{\frac{\pi}{2}(d/2)^4 G_{BC}}$ (3b)

Substituting Eqs. (3a) and (3b) into Eq. (2), we get:

$$\frac{T_A L_{AB}}{\frac{\pi}{2}(\frac{D}{2})^4 G_{AB}} = \frac{T_C L_{BC}}{\frac{\pi}{2}(\frac{d}{2})^4 G_{BC}}$$

$T_A = \frac{D^4 L_{BC} G_{AB}}{d^4 L_{AB} G_{BC}} T_C$ (4a)

Since $L_{AB} = 2$ m, $L_{BC} = 1$ m, $G_{AB} = 40$ GPa, $G_{BC} = 80$ GPa, $D = 0.05$ m, $d = 0.025$ m we obtain:

$T_A = 4T_C$ (4b)

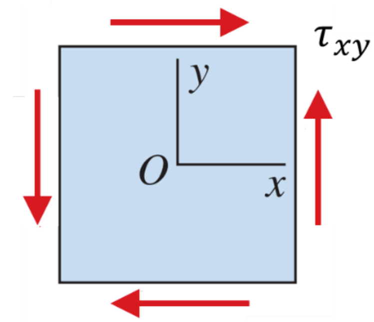
Substituting Eq. (4b) into Eq. (1) we get:

$T_A = 0.8T$ (5a)

$T_C = 0.2T$ (5b)

The stress-state of the surface point H is pure shear (see top figure). The Mohr's circle can be then drawn out (see bottom figure). The two principal stresses can be then determined from the Mohr's circle:

$\sigma_1 = -\sigma_2 = \tau_{\max}$ (6a)



Using torsion formula, for segment AB

$$\tau_{\max} = \frac{T_A D_{AB}/2}{J_{AB}} = \frac{T_A D_{AB}/2}{\frac{\pi}{2}(\frac{D_{AB}}{2})^4} = \frac{16T_A}{\pi D_{AB}^3} \quad (6b)$$

Substituting Eq. (6b) into Eq. (6a), we get

$$|\sigma_1| = |\sigma_2| = \frac{16T_A}{\pi D_{AB}^3}$$

Substituting Eq. (5a) into it, we can get

$$|\sigma_1| = \frac{16}{\pi D_{AB}^3} \times 0.8T$$

$$100 \times 10^6 = \frac{16}{\pi \times 0.05^3} \times 0.8 \times T$$

$$\boxed{T_{\max} = 3068\ \mathrm{N \cdot m}}$$